CUSTOMER CHURN ANALYSIS REPORT

Telecom Churn Prediction — Data Science PoC

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1. Executive Summary

This report presents findings from a customer churn analysis of a telecom dataset comprising 5,000 customers and 19 attributes. The objective was to build predictive models, identify key churn drivers, and deliver actionable retention strategies. Key findings include:

- The churn rate is 18.1%, creating a significant class imbalance addressed via SMOTE oversampling.
- Four models were evaluated: Logistic Regression, Random Forest, XGBoost, and a PyTorch Deep Learning network. All achieved ROC–AUC of ~0.75–0.76, but XGBoost and Deep Learning lead in churn recall (0.82).
- Logistic Regression achieves the highest Average Precision (AP = 0.392), making it preferable when false-positive costs are high. XGBoost is recommended for maximum churn capture.
- Precision remains low across all models (~0.29–0.31), meaning ~70% of flagged customers are false positives — a critical cost factor when designing retention campaigns.
- Top churn drivers: month-to-month contracts, short tenure, elevated support tickets, high monthly charges, Fiber Optic internet service, and electronic check payment method.

2. Model Performance Comparison

The table below summarizes test-set performance (1,000 samples) for all churn-class metrics:

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.66	0.31	0.74	0.437	0.756
Random Forest	0.62	0.30	0.80	0.432	0.755
XGBoost	**0.61**	**0.29**	**0.82**	**0.433**	**0.756**
Deep Learning (PyTorch)	0.60	0.29	0.82	0.4319	0.753

Analysis: All four models show similar ROC-AUCs (~0.75–0.76), indicating comparable overall ranking ability. However, ROC-AUC can be misleading in imbalanced classification problems because it is heavily influenced by the majority class (non-churners). The Precision–Recall perspective reveals that Logistic Regression achieves the highest Average Precision (AP = 0.392), followed by Random Forest (AP = 0.382) and XGBoost (AP = 0.372). XGBoost and the Deep Learning model share the highest recall (0.82), meaning they capture the most churners. If the primary business objective is to maximise churn capture, even at the expense of more false positives, XGBoost is recommended as the final production model due to its superior recall, lower complexity than deep learning, and excellent interpretability.

Recommended Model: XGBoost — delivers the highest churn recall (0.82) with feature-importance interpretability and lower infrastructure cost than the Deep Learning model. For precision-sensitive, budget-constrained campaigns, Logistic Regression (AP = 0.392) offers a better trade-off between precision and recall.

3. Insights

3.1 Data Preparation & Feature Engineering

- Dataset: 5,000 customers, 19 raw features. Missing values in internet_service (14.78%), total_charges (4%), and data_usage_gb (2%) were handled via categorical fill ('None') and median imputation.
- Class imbalance (18.1% churn vs. 81.9% non-churn) was addressed with SMOTE oversampling during training to prevent model bias toward the majority class.
- Feature Engineering — Three domain-driven variables were created to improve predictive signal:
  • tenure_months_group — Bins tenure into lifecycle stages: 'new' (0–12 mo), 'early' (13–24), 'mid' (25–48), 'loyal' (49+). Rationale: churn risk is non-linear with tenure; early-stage customers churn at disproportionately higher rates, and

encoding this as a categorical feature helps both linear and tree-based models.

• services_count — Sum of add-on services (streaming TV, tech support, online backup). Rationale: higher service adoption indicates deeper platform engagement and 'stickiness'; customers with zero add-ons are significantly more likely to leave.

• high_risk_flag — Binary flag that activates when a customer is on a month-to-month contract, pays via electronic check, AND has zero add-on services. Rationale: This three-way combination captures the highest-risk behavioural profile — minimal commitment, frictionless payment exit, and low engagement — identified during exploratory data analysis.

3.2 Key Churn Drivers

- Contract Type: Month-to-month customers churn at significantly higher rates than those on annual or two-year contracts.
- Tenure: Customers with < 12 months tenure are the highest-risk group; early onboarding experience is critical.
- Monthly Charges: Higher charges correlate with churn, especially among customers who do not perceive proportional value.
- Support Tickets: Elevated ticket counts are a strong churn signal, reflecting ongoing service dissatisfaction.
- Internet Service: Fiber Optic customers churn more than DSL or no-internet customers, likely due to pricing or quality gaps.
- Payment Method: Electronic check users churn more — possibly reflecting lower commitment compared to auto-pay methods.

3.3 High-Risk Customer Segments

- Short-tenure (< 12 months), month-to-month contract customers with Fiber Optic internet and no tech support or online backup.
- Customers with above-median monthly charges (> ~$64.60) and 3+ support tickets in the last quarter.
- Senior citizens without partners, particularly those using electronic check payment.

4. Business Recommendations

Based on the analysis, we recommend the following three strategies to reduce customer churn:

Recommendation 1: Proactive Retention for Short-Tenure Customers

- Insight: Short-tenure customers (< 12 months) on month-to-month contracts are the highest-risk segment.
- Action: Implement a structured onboarding programme with personalised check-ins at 30, 60, and 90 days. Offer incentive-based upgrades to annual contracts within the first quarter.
- Impact: Targeting early-stage churn could reduce attrition by 15–20%; retaining existing customers costs 5–7× less than acquiring new ones.

Recommendation 2: Improve Fiber Optic Service Quality & Support

- Insight: Fiber Optic customers churn at higher rates, and elevated support tickets are a strong churn predictor.
- Action: Conduct root-cause analysis of Fiber Optic complaints. Create a dedicated support tier for high-value Fiber customers and proactively reach out when tickets exceed 3 in a 3-month window.
- Impact: Addressing the core dissatisfaction drivers could reduce Fiber Optic churn by 10–15%.

Recommendation 3: Drive Service Bundling & Value-Based Pricing

- Insight: Customers with zero add-on services (tech support, streaming, online backup) churn at significantly higher rates. High monthly charges without perceived value further accelerate attrition.
- Action: Introduce discounted service bundles that encourage adoption of 2+ add-ons. Offer value-aligned pricing tiers, so high-charge customers receive visible benefits. Incentivize migration from electronic checks to auto-pay methods (e.g., small monthly discount) to increase payment stickiness.
- Impact: Increasing service adoption deepens engagement and raises switching costs. Customers with 2+ services show markedly lower churn, making bundling a high-leverage retention lever.